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13
A Novel Approach for Object Detection Technique Using Deep Learning

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Abstract

This automation era has seen a rapid expansion of the fields of computer vision and image processing. Applications for object detection include face detection, video surveillance, autonomous driving, image retrieval, pedestrian identification, human behavior analysis, and face photo tagging. The task of recognizing and categorizing everything that appears in images or videos is known as object detection (also known as object localization). Machine learning (ML) is a branch of artificial intelligence (AI) that focuses on developing models and algorithms that let computer systems carry out particular functions. Feature extraction and classification are the two primary steps that often comprise machine learning tech-

niques in the context of picture classification. The photos are processed using feature extraction techniques including histogram of oriented gradients (HOG), scale-invariant feature transform (SIFT), and Haar-like features to retrieve pertinent data. Classifiers like Adaboost, deformable part models (DPMs), and support vector machines (SVMs) use these attributes to classify the images. There are many reasons why object detection is difficult, such as uneven lighting, occlusions, different placements, small datasets, and imbalances in the classes. Because deep learning techniques were used to further automate the feature extraction procedure, the performance has increased. Issues including occlusions, position and perspective changes, object rotation and scale changes, different object sizes, distortion, and intraclass variance remain unresolved even though convolutional neural networks (CNNs) are currently being researched. In different applications, object detection algorithms are impacted by each of these aspects. The most recent developments in Deep Learning techniques for object detection and identification are examined in this article. These developments have enormous promise for the fields of image processing and computer vision. The goal of this study is to present various methods that, when compared to the existing approaches, may perform Object Detection with a significantly greater level of accuracy and reasonable real-time performance. One of the primary problems with object recognition is image classification, or predicting the class of a picture. When an image contains an object collection, image localization becomes a little trickier to solve. The class and location of the objects throughout the image should be the system's prediction in this instance. The most difficult aspect of the object detection task is combining localization and categorization. The present study only considers the VGG-16 model, despite the fact that there are several deep learning models for object detection accessible in the literature. The study that is now being published indicates that for more complex applications, the accuracy of these models in detecting objects needs to be further improved.

Therefore, the current research effort aims to achieve a significant improvement in object detection accuracy above the current levels. This study starts with the fundamental ideas of deep learning before delving further into various deep learning architectures. The four hyper-parameters that affect object detection accuracy—"learning rate," "epoch rate,"

"training batch size," and "testing batch size"—are the subject of the current work. The learning rate of a model is the speed at which it chooses weights from the neural network.

Keywords: Neural network, computer vision, deep learning, feature extraction, learning rate, object event detection

13.1 Introduction

When it comes to many applications including image retrieval, video surveillance, pedestrian identification, automated face tagging, picture classification, and human behaviors, object detection is widely used and helpful in comprehending images that are used in Robot vision, autonomous driving, facial recognition, and analysis. Deep learning techniques, which are employed in various object detection applications, have been made successful by the availability of big datasets and GPUs.

13.1.1 Object Detection Techniques

One of the most used methods for object detection is the sliding window algorithm. This method uses a convolution neural network to identify items in an image. This technique slides windows on the image matrix, each with a variable size. Despite producing results that are passably decent, this method's significant computational cost is a result of its enormous number of windows. Convolutional modeled sliding window detection is one approach to solving this issue.

A CNN approach for object identification is shown in Figure 13.1. With 16 filter matrices, each measuring 5 by 5, the original 14 * 14 * 3 image matrix was produced. Following the process of convolution, the picture matrix's scale was 10 * 10 * 16. The picture matrix's size is then further decreased to 5 * 5 * 16 by applying the maximum pooling technique to it. The softmax layer, which ascertains the object's class, comes after a completely linked layer and this matrix. Four categories of objects are categorized in Figure 13.1.

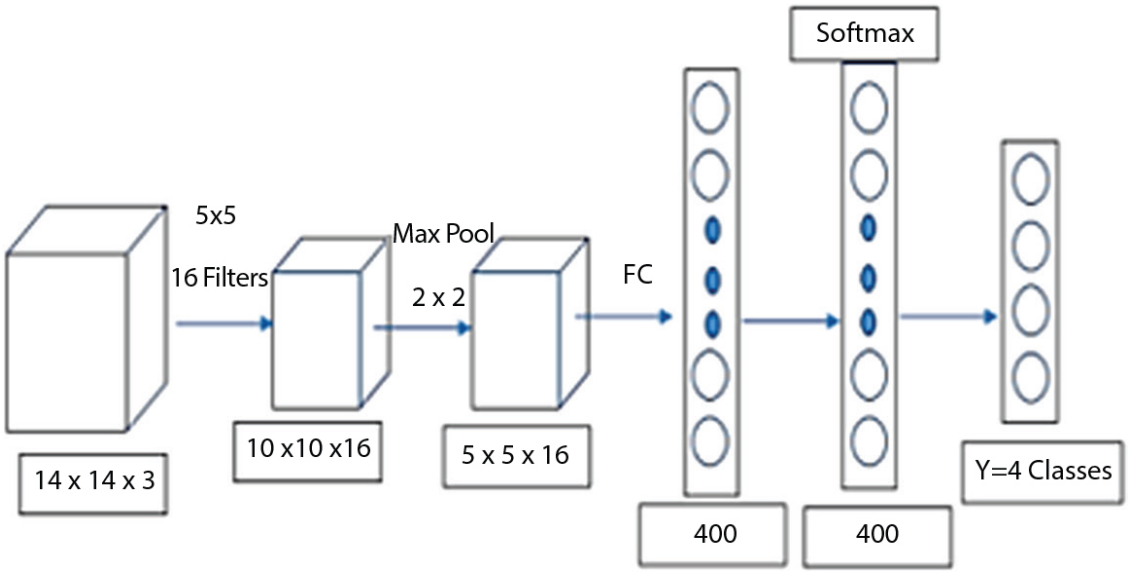


Figure 13.1 CNN layer.

The accuracy of Deep Learning Models' image recognition makes them a popular choice for Object Detection Algorithms. Convolution Neural Network Architectures have advanced, and new algorithms have been implemented, leading to advances in Deep Learning. Among machine learning algorithms, deep learning [14] is a subset that uses weights to learn features. For the purpose of classifying the objects seen in images and videos, these models take out specific traits. These DLMs have a wide range of uses, including voice recognition, biomedical image analysis, cyber security, renewable energy generation, weather forecasting, video object analysis, image processing, biometric recognition, iris recognition, and natural language processing. Utilizing many layers of artificial neurons, these models implement the convolutional neural network (CNN) idea.

13.1.2 Deep Learning-Based Object Detection

Machine learning's subset of deep learning uses data to identify features. With the expansion of information, Deep Learning algorithms outperformed Machine Learning algorithms in terms of efficiency. By effectively training neural networks, "Geoffrey Hinton" made a significant contribution.

Reducing training time by using the pre-trained classification network as the base network and building an end-to-end deep learning-based object detector is one benefit of the deep learning-based object detection tech-

nique. The first step in object detection using a deep learning framework is to choose a framework from the literature, such as a faster region-based convolutional neural network (R-CNN), single shot detector (SSD), and you only look once (YOLO) [15]. Next, these frameworks are executed on a suitable base network.

13.2 Literature Survey

An innovative approach to vehicle instance segmentation was developed by Lichao Mou *et al.* [5]. Pixel by pixel, the positions of the cars are located, and each pixel is linked to a real-world instance of a vehicle. This work combines two complementing tasks into a single multitask learning network: semantic boundary detection and vehicle region segmentation. Additionally, a new dataset called busy parking lot unmanned aerial vehicle video has been created for use in future studies on vehicle instance segmentation.

A pixel classification method utilizing segmentation methodology to identify stationary items in the image was proposed by Waqas Hassan *et al.* [8]. Next, these objects are monitored with an innovative method known as adaptive edge orientation. Compared to state-of-the-art techniques, this technology recognizes objects with a 95% accuracy rate.

The justifications for CNN use in computer-aided detection issues were covered by Hoo-Chang Shin *et al.* [12]. This research focuses on two specific computer-assisted detection tasks: identifying interstitial lung disease (ILD) and spotting thoraco-abdominal lymph nodes (LNs).

Xuesong Zhang *et al.* [13] introduced a method for combining several fine-tuned CNN models to classify objects into multiple categories. Using this technique, the final fully connected layers of the Fine-tuned CNN models are modified based on the intended task after they have been trained on the target dataset. The suggested method produces results that are adequate for object detection.

Various frameworks for object detection based on deep learning are discussed by Zhao *et al.* [14]. This work focuses on optimizations that can be

applied to object detection architectures to further boost their efficiency. Additionally, this effort provided several encouraging avenues for improving the object detector design.

The training technique proposed by Olaf Ronneberger *et al.* [16] makes considerable use of the existing images through data augmentation. This method performed even better than the previous top performer in the ISBI challenge for segmenting neural structures in electron microscopic stacks, even when trained with just a small number of images.

A rotation-invariant approach to the detection of geographical objects from satellite pictures was proposed by Yongtao Yu *et al.* [17]. Particularly in the suggested method, three techniques were applied. In order to produce non-redundant patches, a super pixel segmentation technique was first proposed. To generate high-level feature representations of the patches, a multilayer deep feature generation model was constructed. Then, to cast rotation-invariant votes for object centroids estimation, a set of multi-scale Hough forests with embedded patch orientations was built. When compared to cutting-edge techniques, this strategy produced excellent outcomes.

Wei Shen *et al.* [18] presented a unique fully convolutional neural network with many side outputs correlated with scale. By taking into account the relationship between the scales of the skeletons acquired by each layer's receptive field widths, the network integrates two scale-associated side outputs at each step. Multi-task learning is used to train the network on two distinct tasks: skeleton localization, which determines if a pixel is a skeleton or not, and predicting skeleton scale, which entails estimating the scale of each individual skeleton pixel. At certain phases, the ground-truth skeletons get supervision from the scale-associated side outputs that are directed toward them at the relevant scales. Then, the responses of the side outputs associated with each scale are merged in a scale-specific way to efficiently detect skeletal pixels across several scales. This approach was evaluated using two skeleton extraction datasets, and the results were promising.

Xiaoning Han *et al.* [19] introduced a unique dueling architecture deep Q-learning network (DQN). This network effectively integrates the predic-

tions of action type and action range, facilitating the identification of the action space. The findings of numerous approaches' thorough testing indicate that the suggested approach produced the best outcomes and accurately predicted the action in real time.

Kwang-Ju Kim *et al.* [20] proposed an innovative method of object detection that adds more prediction layers to the traditional Yolo-v3 by using spatial pyramid pooling. This strategy improves the detection of occluded vehicles with a 24% increase in accuracy when compared to industry-leading algorithms utilizing the UA-DETRAC dataset, including DPM, ACF, RCN, compact, NANO, EB, GP-FRCNN, SA-FRCNN, faster-R CNN2, HAVD, and SSD-VDIG.

Jing Liu *et al.* [23] created the multi-component fusion network (MCFN), a deep convolutional neural network (DCNN), to increase the precision of small item detection. The researchers developed a dual pyramid fusion network that integrates semantic and spatial data in order to capture the characteristics of small objects. After extracting features of small object samples with an area proposal network, contextual information is integrated. The outcomes demonstrated that this approach worked better than cutting-edge methods, particularly in challenging situations involving occlusion.

13.3 Deep Learning Methods

This illustrates various Deep Learning methods used in this study for Object Detection and Localization.

13.3.1 CAFFE Object Detection System

A deep learning system called convolution architecture for feature extraction (CAFFE) [26] was developed by Yangqing Jia and Evan Shelhamer. The CAFFE object detection system acquired popularity due to its quick processing performance. Using the CAFFE system for object detection has several benefits, including speed, modularization, openness, and quick start. A network is defined by CAFFE as an assembly of layers. There are

two components to CAFFE layers. Layer parameters and Layer properties are what they are depicted in Figures 13.2 and 13.3.

Layer type, layer name, and layer connection structure are all contained in layer properties. CAFFE uses blobs to convey all of its information. A unified memory interface, or blob, is a type of standard array. The array utilized for data processing is N-dimensional. It offers the synchronization of the GPU with the CPU. The information saved and transferred across layers and nets is described in depth by the blob. Blobs use the "Synced Mem" class to synchronize the CPU host and GPU device, hiding the mental and computational complexity of CPU/GPU operation. Batches of photos, model parameters, and optimization derivatives are a few examples of blobs. The different convolution layer parameters, like Kernel size and Stride, are referred to as Layer Parameters. When an image is run through the CAFFE Object Detection system, these settings are required in order for the image to learn features.

Layer Properties
Conv1 “name”
Type: CONVOLUTION
Data “bottom”
Conv1 “top”

Figure 13.2 CAFFE layer properties [1–11].

Layer Parameters
Convolution_param{
Num_output:10
Kernel_size:3
Stride:1
Weight-filler{
<span "style= color:" #c0504d
type:; ">xavier"}}

Figure 13.3 CAFÉ layer parameter [1–11].

The data layer, convolution layer, pooling layer, activation functions, loss functions, and dropout are some of the parts that make up CAFFE’s basic computational unit. After starting from zero to learn the model, it continues to learn from the saved models. Next, in order to identify new things, it uses a process known as “fine-tuning.”

The CAFFE model uses the “CAFFE train” command to train the network; moreover, a solver configuration file called “solver.prototxt” is required.

This file has information about network training. The CAFFE model uses the "bvlc_reference:CAFFEnet" as its foundational network to construct the Object Detection framework through the use of "transfer learning." The skill of creating a model for one job and applying it to another is known as transfer learning. The ImageNet dataset [27], which has millions of photos in 1000 categories, was used to train the CAFFE model. In Figure 13.4, the picture is examined by using the CAFFE object detection system.

13.3.2 Keras Object Detection Model

Carbon Francois Chollet created and oversaw the development of Keras [28]. Using this high-level library, neural network models can be constructed (Figure 13.5).

It is very flexible interface makes it straightforward to develop even more complex neural network models. The capacity of Keras to execute ideas quickly is by far its greatest advantage.

This paradigm makes use of three principles: extensibility, minimalism, and modularity. A model's modularity can be understood as either a sequence or a graph on its own. Whereas minimalism means that the Keras Library gives just enough to generate an outcome that has no frills and maximum readability, all of the concerns of the deep learning model are distinct components that can be merged freely. Extensibility is the straightforward insertion of new elements that enable researchers to assess and formulate novel ideas.

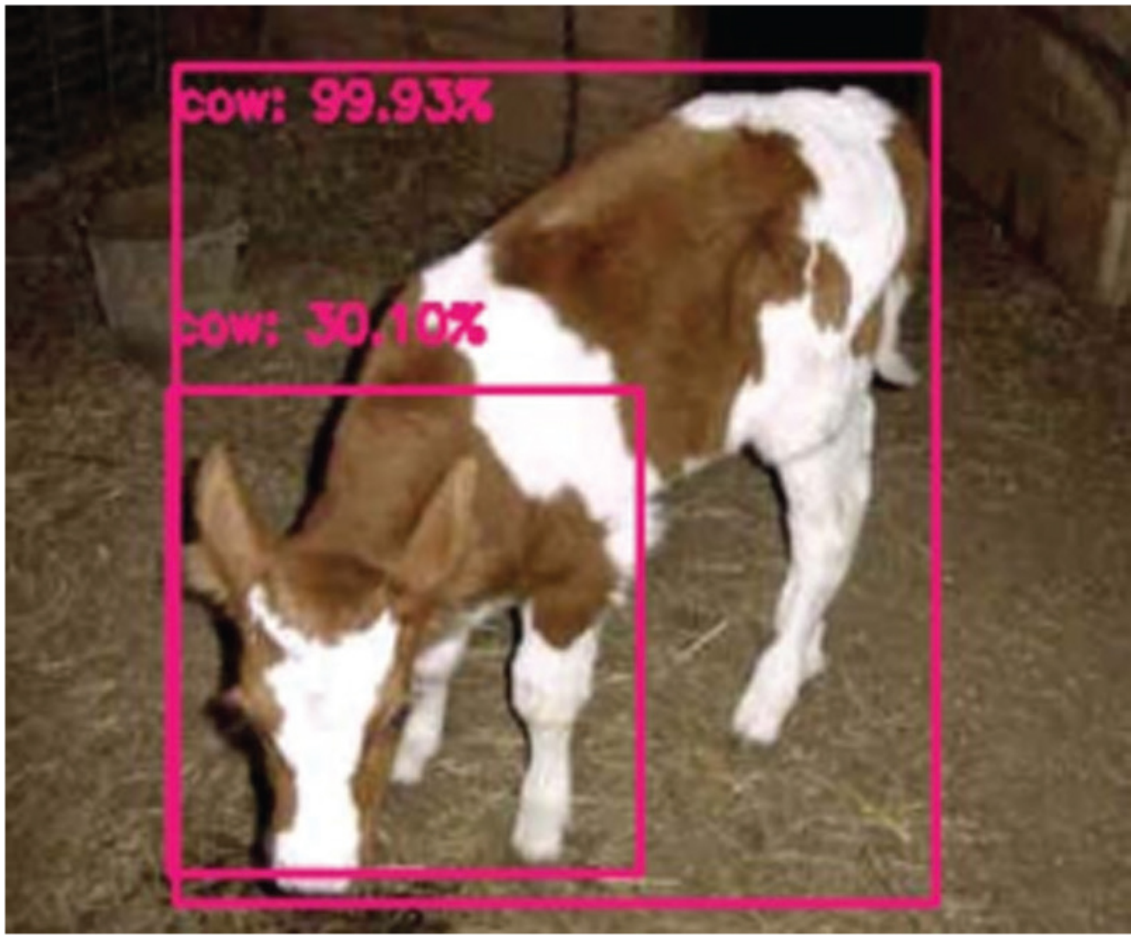


Figure 13.4 CAFÉ object detection example [1–11].

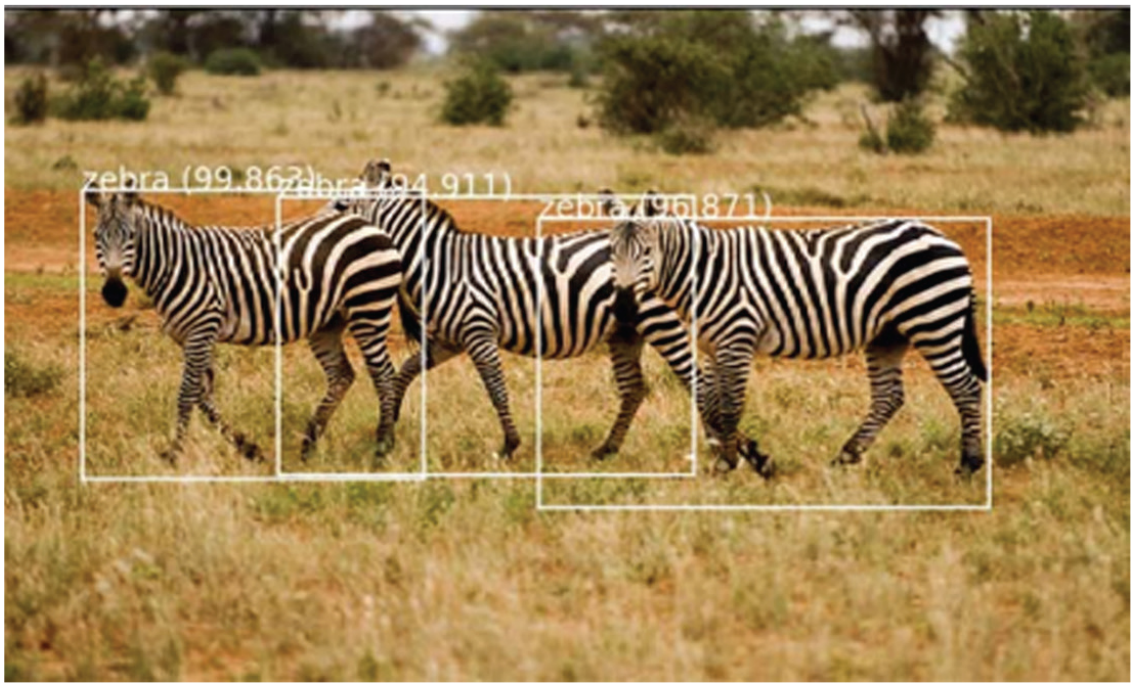


Figure 13.5 Keras object detection [1–11].

An open-source Python library called Keras is used to create and run deep learning models. It is made up of libraries for numerical computation like tensor flow and Theano, which use very few lines of code to train neural networks. A series of layers is what defines Keras models. These

models create deep learning models by following six important phases. After loading the necessary data and creating the neural network in Keras, they build the model, train it on the data, assess it on the data, and use it to generate predictions.

Table 13.1 Comparison of the object detection system.

S. no.	Model name	Base network	Number of layer	Error rate (%)
1	CAFFE model	Alex Net, VGGNet 16/17	6 Layers	84.5%
2	Keras	Inception-V3	160 Layers	94.5%

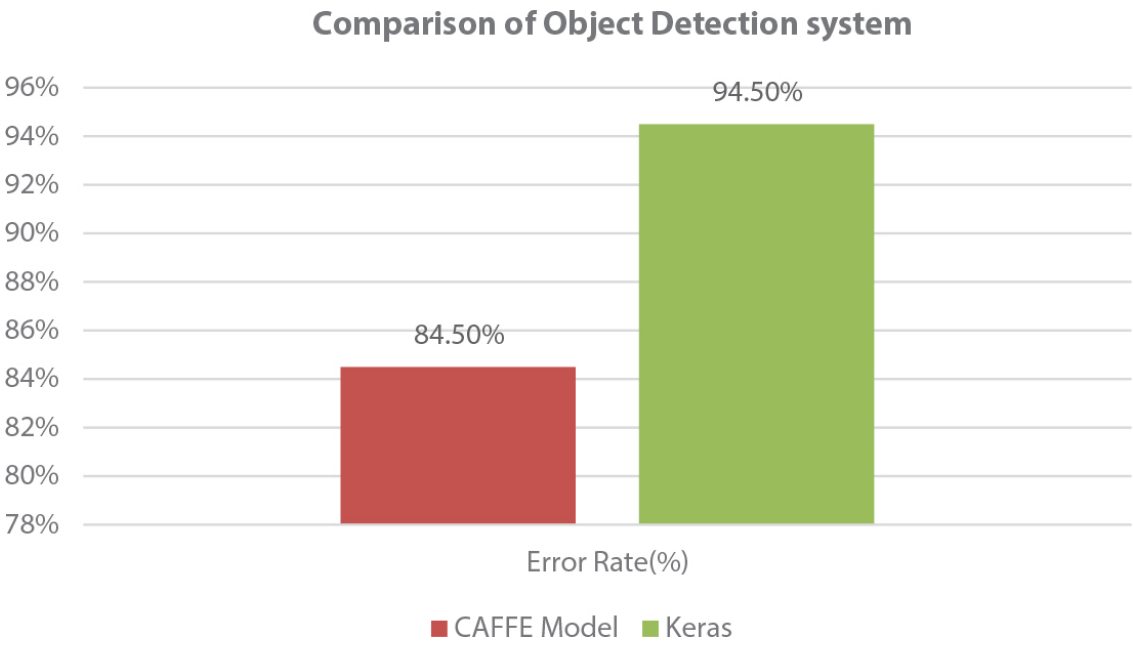


Figure 13.6 Comparison of the object detection system graph [1–11].

Table 13.1 provides an overview of each deep learning system, including information on base networks, layer count, and error rates. The percentage of times the classifier failed to determine the right class of the object in the image among its first five choices is known as the error rate (Figure 13.6).

13.4 Deep Learning Models

One type of network that incorporates a novel training method is called a deep learning network. For the purpose of processing photos, it is a multi-layer network. These methods are used in deep learning models that are described in below:

13.4.1 Visual Geometry Group-16 (VGG-16)

In their research paper titled "Very Deep Convolutional Networks for Large-Scale Image Recognition," K. Simonyan and A. Zisserman suggested the Visual Geometry Group (VGG-16) as a convolutional neural network [29]. Figure 13.7 illustrates the architecture of the VGG-16 DLM, which consists of sixteen convolution layers, three fully connected layers, five maximum pooling levels, and a softmax layer acting as the last layer for image categorization.

Every hidden layer receives an application of rectified linear unit (ReLU) activation functions. A 224 x 224 RGB image is the input for the first convolution layer. This model uses filters with a stride of 1*1 pixels and a pixel size of 3*3. With a stride of 2, maximum pooling is carried out over 2*2 pixels. At the conclusion of the network are layers that are fully connected.

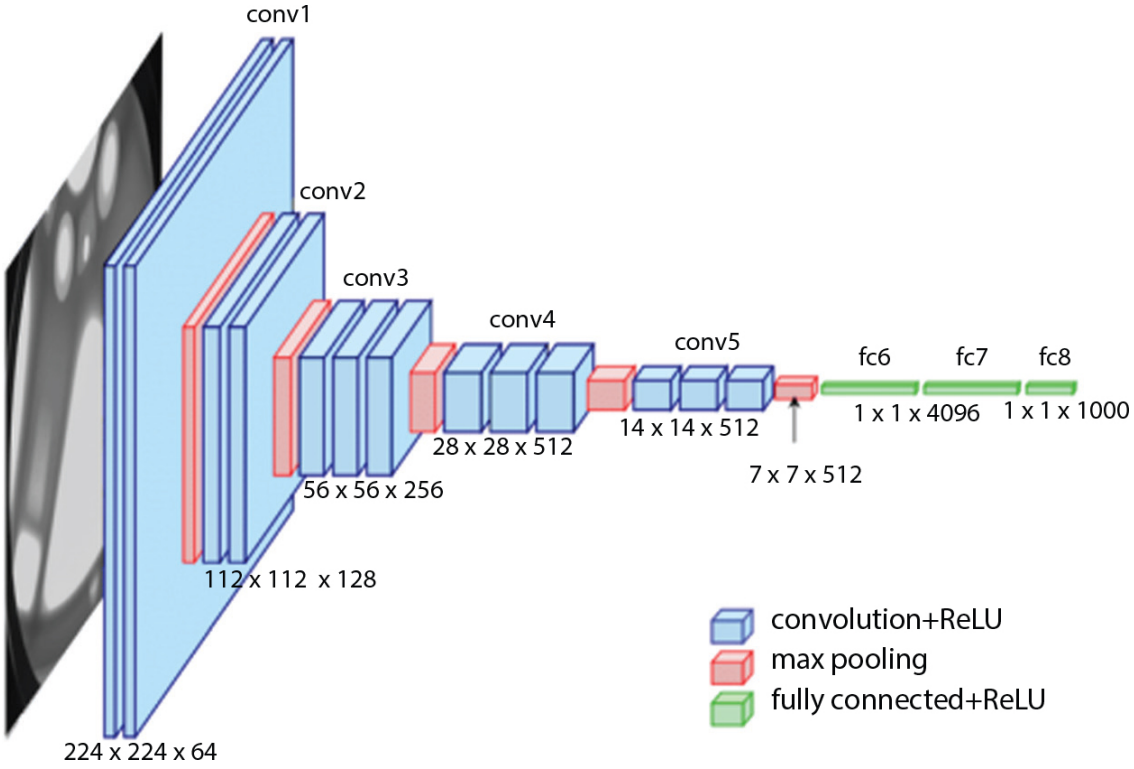


Figure 13.7 VGG 16 architecture.

The system is made up of three fully connected layers: the first two having 4096 channels each softmax layer sits at the bottom, with 1000 channels. With 1000 channels in the last layer, the model can detect 1000 things. This model has 138 million parameters overall and weighs 533 MB. This model achieves 70.5% accuracy on the Imagenet Dataset.

13.4.2 Visual Geometry Group-19 (VGG-19)

VGG-19 is an improved VGG-16 variant that needs a picture that is 224 x 224. Figure 13.8 shows the architecture of VGG-19, which when combined with the ImageNet Dataset, is capable of classifying images into 1000 different categories. It is made up of 19 deep network layers. This network increases depth by stacking three 3*3 convolutional layers on top of each other.

This model has 55144 million parameters and 549 MB in size for the VGG-19 model. Using the ImageNet Dataset, this model has a 75.2% accuracy rate.

13.4.3 Inception

In their 2014 publication "Going Deeper with Convolutions" [30], a group of scientists led by Christian Szegedy, Wei Liu, Yangqing Jia, Scott Reed, Dragomir Anguelov, Dumitru Erhan, Vincent Vanhoucke, and Andrew Rabinovich presented the idea of inception. An RGB image with dimensions of 299x299 is fed into the convolution layer of the Inception model. There are 27 layers in the Inception-V1, the first iteration of the Inception model.

After incorporating the batch normalization technique, the model was renamed InceptionV2. Subsequently, additional layers and factorization were introduced, resulting in a revised version known as Inception-V3, which consists of 48 layers instead of the previous 27. It is important to remember that the Inception-V1 model has almost 7 million parameters.

The model has 24 million parameters and a 79.5 MB file size. With varying numbers of convolution, pooling, dropout, and softmax layers, the Inception-V3 model's architecture is depicted in Figure 13.9. Utilizing the ImageNet Dataset, this model achieves 78.8% accuracy.

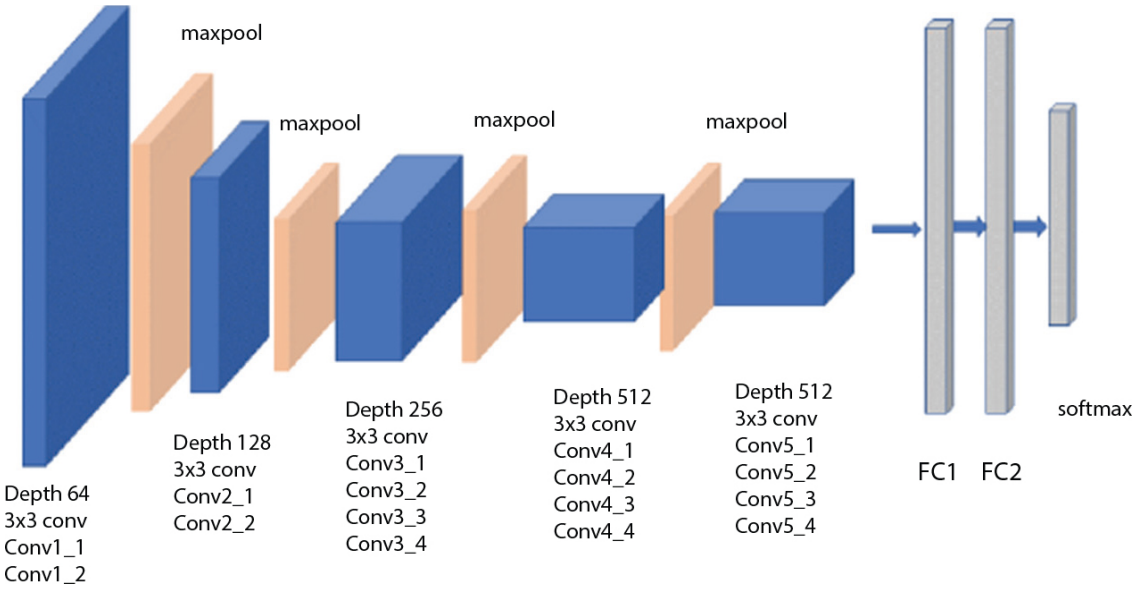


Figure 13.8 VGG-19 architecture.

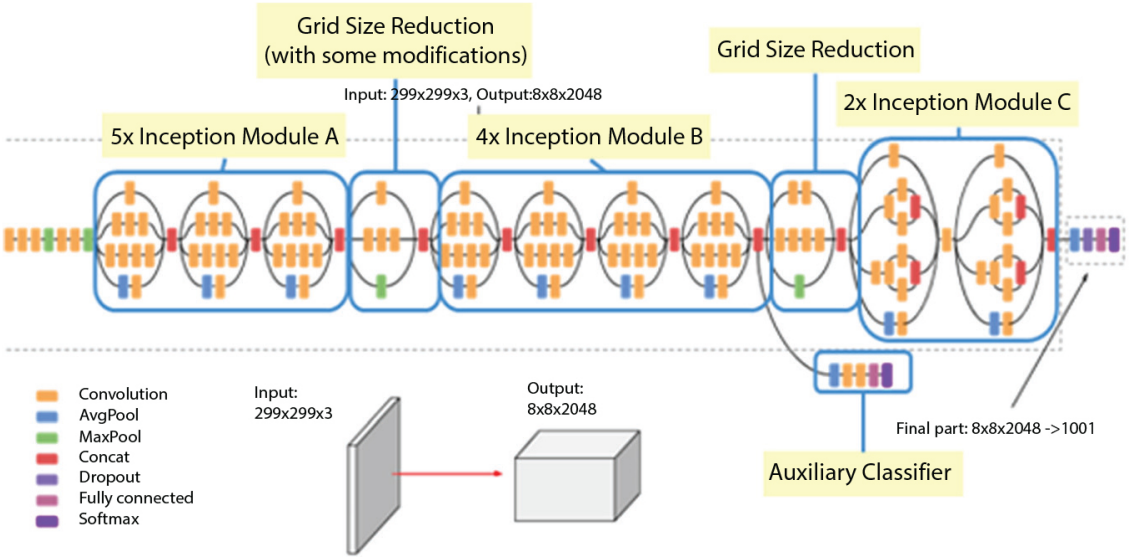


Figure 13.9 Inception architecture.

13.4.4 Number of Epochs

The number of times the learning algorithm runs through the whole training sample dataset is referred to as the "Number of Epochs" [31]. The forward and backward passes of a training sample make up an epoch. The training dataset sample adjusts the neural network's weights following each epoch, pointing out that there are almost 7 million parameters in the Inception-V1 model.

13.4.5 Activation Functions

Using these functions, inputs and outputs can be mapped in a complicated functional way. During the conversion of input signals into output sig-

nals, they add nonlinearity to the network. By learning complex input, including voice, video, and images [32], these functions primarily aim to increase the network's power. Tanh, ReLU, and Sigmoid are a few of the most often-used activation functions. An activation function that is affixed to a CNN neuron is seen in Figure 13.10.

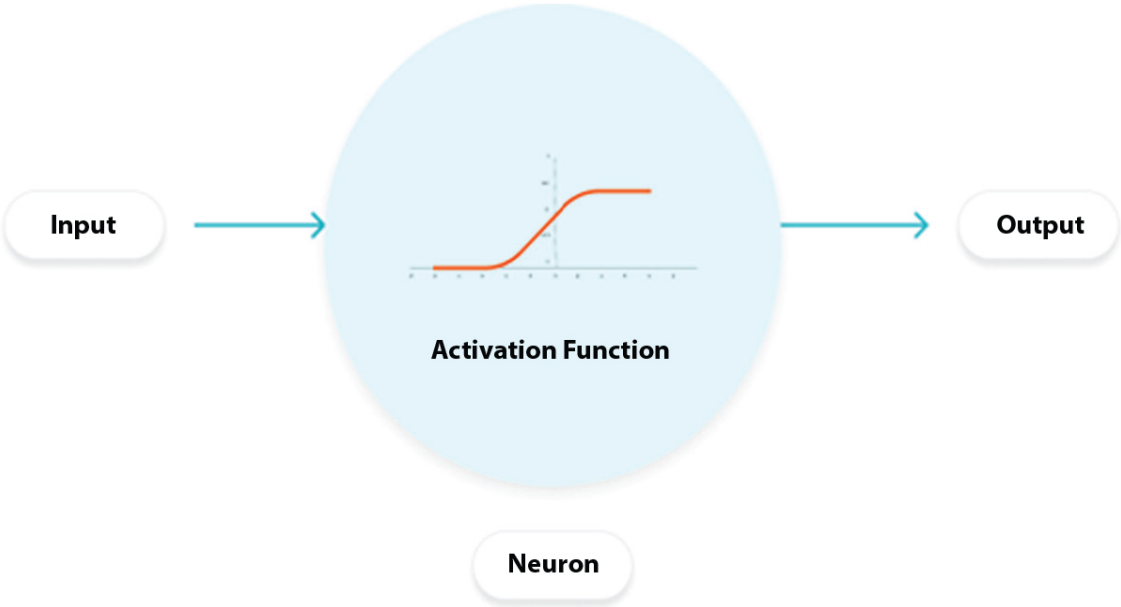


Figure 13.10 Activation function of a neural network.

Evaluating a neural network's performance is done by mathematical formulas called activation functions. All of the network's neurons have these functions linked to them, which decide whether or not to activate them depending on how the input relates to the model's prediction.

13.5 Experimental Results

The experiment used test photographs, and the findings show that as the learning rate drops, object detection accuracy varies. Moreover, the results show that the percentage of incorrect guesses falls as prediction accuracy rises.

A Table 13.2 summarizing the experimental results shows that a learning rate of 0.4 minimizes the amount of false predictions while achieving a maximum object detection accuracy of 91.05%. This performed better than the existing literature's state-of-the-art accuracy of 71.4%, as clearly visualized in Figure 13.11.

The results show that when VGG-19 DLM is additionally implemented at different learning rate values, the highest object detection accuracy is attained at 92.86% and the number of wrong predictions is capped at learning rate = 0.4. This also performed better than the published state-of-the-art accuracy of 75.2%.

Table 13.2 Comparison of the object detection system [21–25].

S. no.	Learning rate	Prediction accuracy (%)	Wrong prediction
1	0.1	2.00	936
2	0.2	1.02	945
3	0.3	1.01	944
4	0.4	91.05	78
5	0.5	71.40	280
6	0.6	71.40	280

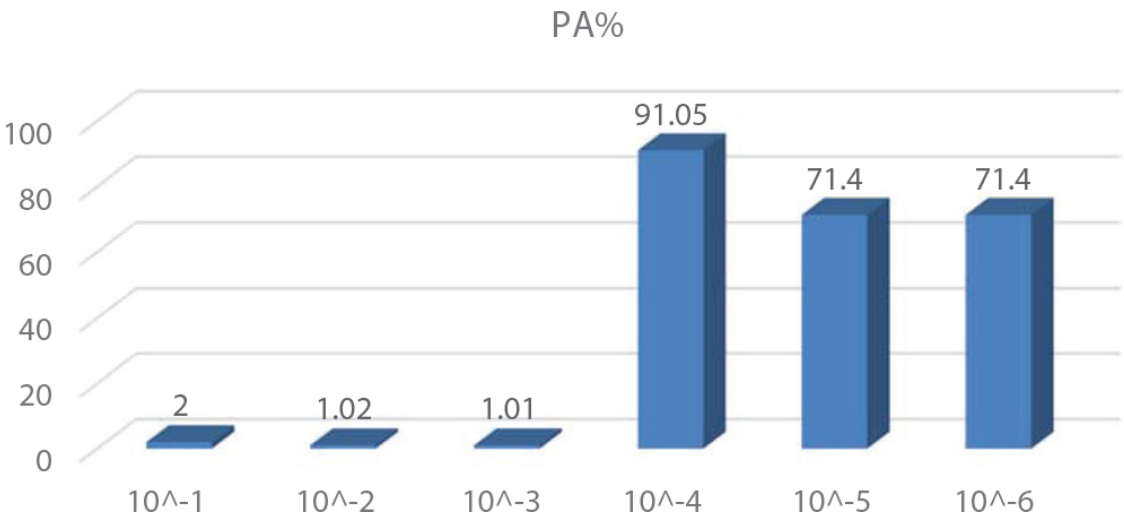


Figure 13.11 Activation function of a neural network [21–25].

Table 13.3 and Figure 13.12 are used to compare the Inception-V3 model with its predecessors, which have more hidden layers. For the model's

designated dataset, the object detection accuracy is assessed. According to the results, accuracy can be improved at a learning rate of 10^-5, and there are very few incorrect predictions.

The Inception-V3 model's experimental results are displayed in Table 13.4. According to these results, the maximum object detection accuracy is 93.59%, as illustrated by the graph in Figure 13.13, with only 62 false positives. These results also surpassed the previous prediction accuracy of 78.5%.

Table 13.3 Comparison of the object detection system.

S. no.	Learning rate	Prediction accuracy (%)	Wrong prediction
1	0.1	1.02	958
2	0.2	1.02	958
3	0.3	1.02	958
4	0.4	92.86	69
5	0.5	67.30	314
6	0.6	7.65	893

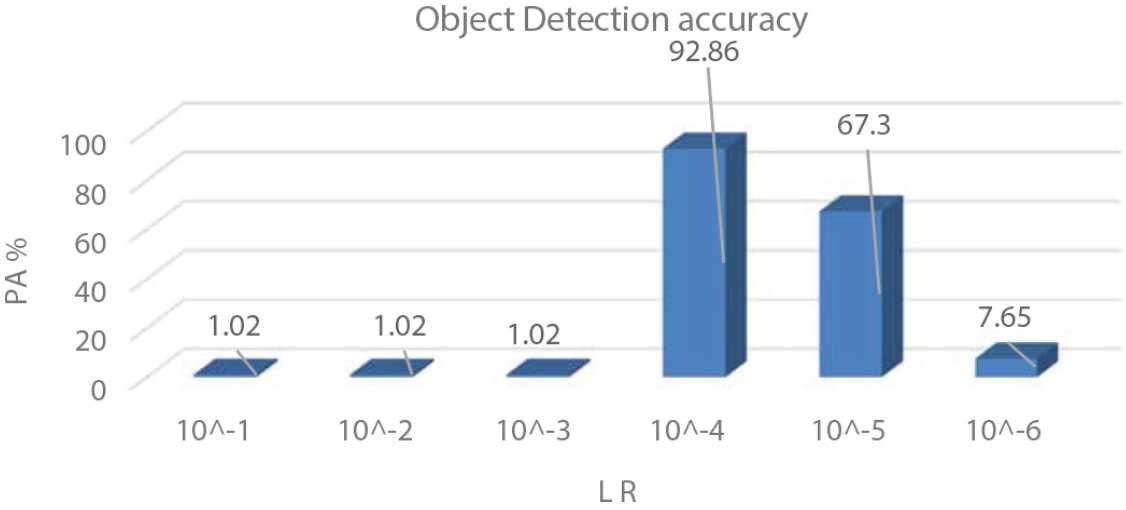


Figure 13.12 Object detection accuracy graph [21–25].

Table 13.4 Inception-V3 accuracy for various learning rates.

S. no.	Learning rate	Prediction accuracy (%)	Wrong prediction
1	0.1	2.06	945
2	0.2	1.85	947
3	0.3	1.85	947
4	0.4	91.80	79
5	0.5	93.59	62
6	0.6	84.05	155

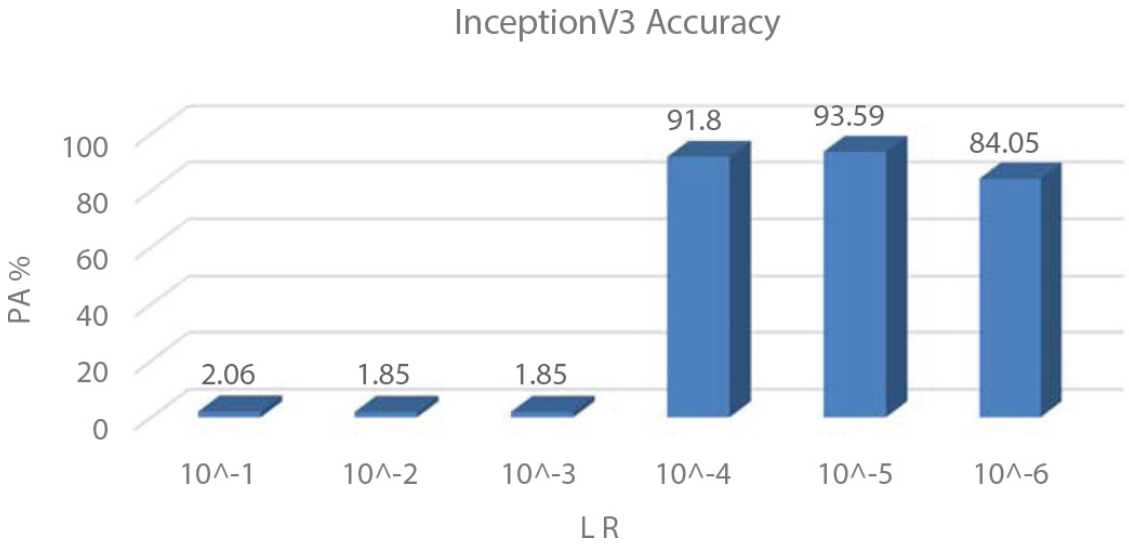


Figure 13.13 Object detection accuracy graph [21–25].

13.6 Conclusion and Future Scope

Increased detection accuracy is needed in the quickly evolving field of object detection. The government invests trillions of dollars yearly in areas like artificial intelligence development, medical science and development, and national security review, among others, where this research area is very helpful. Much research is being done in this area to determine how to increase object detection accuracy. Convolutional neural network is used to find out the maximum accuracy. The three major purposes of the models found in the literature are image classification, object detection, and object recognition. For this purpose, every model has a certain accuracy level. The primary objective of the current study is to increase object detection accuracy.

The accuracy of object detection can be affected by a number of additional parameters that are available in the literature in addition to the ones that are taken into consideration. These include the number of top layers that need to be trained, the optimization function, the drop-out rate, and the activation function. Numerous activation functions can be found in the literature. The optimal activation function for the chosen application can be found by repeating deep learning techniques with several activation functions.

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